Example 2: Use a cofactor expansion across the third row to compute $\det A$, where $A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$

Soln: $C_{31} = (-1)^{3+1}\begin{vmatrix} 5 & 0 \\ 4 & -1 \end{vmatrix} = 5\cdot(-1) - 0\cdot(4) = -5$

$C_{32} = (-1)^{3+2}\begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} = (-1)\cdot(1\cdot(-1) - 0\cdot 2) = 1$

$C_{33} = (-1)^{3+3}\begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix} = 1\cdot 4 - 5\cdot 2 = -6$

By Theorem 1, $\det A = a_{31}\cdot C_{31} + a_{32}\cdot C_{32} + a_{33}\cdot C_{33}$

$= 0\cdot(-5) + (-2)\cdot 1 + 0\cdot(-6) = -2$ ☐

Theorem 1 is helpful for computing the determinant of a matrix that contains many zeros. For example, if a row is mostly zeros, then the cofactor expansion across that row has many terms that are zero, and the cofactors in these terms need not be calculated.

The same approach works with a column that contains many zeros.

Example 3: Compute $\det A$, where

$$A = \begin{bmatrix} 3 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

Soln: By Theorem 1, $\det A = C_{51}\cdot a_{51} + a_{52}\cdot C_{52} + a_{53}\cdot C_{53} + a_{54}\cdot C_{54} + a_{55}\cdot C_{55}$

$a_{51} = a_{52} = a_{53} = a_{55} = 0$.

$\therefore \det A = a_{54}\cdot C_{54} = -2\cdot(-1)^{5+4}\cdot \det \begin{bmatrix} 3 & -7 & 8 & -6 \\ 0 & 2 & -5 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 \end{bmatrix}$

$= 2\det \begin{bmatrix} 3 & -7 & 8 & -6 \\ 0 & 2 & -5 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 \end{bmatrix}$